

Gateway validation report

HealthClaimHardened integration | Gateway-local proof generation and verification

PASS WITH OBSERVATIONS

Decision

Z10 is implemented and local positive/adversarial tests pass. The Gateway creates a HealthClaimHardened Groth16 proof, verifies it before returning an envelope, binds public fields to configured authority/root/policy and request context, and excludes private witness data from JSON. This prototype is not production-ready: trust/replay stores and proof keys are process-local or ephemeral. No Fabric transaction is submitted.

Scope and environment

Item	Observed
Repository	Clean main at start; HEAD 4608955dfac7bf44dd89a139e67ae79dbe81c96d (Z9).
Host	Linux/amd64; AMD Ryzen 7 PRO 8840HS with Radeon 780M Graphics; 27 GiB RAM reported.
Go	Host default 1.22.2; Gateway module selects Go 1.25.8 (declares Go 1.25.7).
ZKP	gnark v0.16.3; gnark-crypto v0.21.0; BN254 Groth16; Z7-v1.
Boundary	No Fabric submit, chaincode, network, Docker or IPFS timing. Existing Fabric and age-proof methods remain.

Existing Gateway architecture

Existing Go code initializes Fabric wallet/network, legacy zerotrust/zkp age-range service, optional encrypted IPFS and contract calls. WriteHealthRecord and ReadHealthRecord remain unchanged. Z10 adds a separate Fabric-independent HealthClaimGateway API in the same module. Legacy zkp1 remains untouched.

Integration and data flow

Stage	Gateway behavior
Challenge	IssueChallenge maps registered recipient/deployment IDs and creates an unpredictable BN254 scalar; lifetime is five minutes in process-local state.
Trust	Generation resolves authority key, accepted root/version, policy commitment/version and identifier mappings through explicit providers. Request fields cannot override trusted root/policy.
Private witness	Caller provides diagnosis, lab/policy/coverage/secret, bounds/set, signature and Merkle path. Canonical field and Boolean path checks precede assignment.
Proof	Builds existing Z7 assignment, generates Groth16 proof, verifies it locally; errors return no successful envelope.
Acceptance	After local verification, consumes challenge and marks record-bound nullifier in memory; state is not shared or durably atomic.
Envelope	Circuit/version, proof bytes, canonical public statement and SHA-256 proof identifier only; no Fabric call.

Input separation

13 public field elements: RegistryRoot, RootVersion, PolicyCommitment, PolicyVersion, ClaimAmount, AuthorityKeyX, AuthorityKeyY, Nullifier, ProtocolDomain, Challenge, RecipientID, DeploymentID, ContextCommitment. Decimal strings are canonical; proof []byte is base64 in JSON.

31 private signals: Diagnosis, LabValue, PolicyID, CoverageCeiling, PatientSecret, LabMin, LabMax, PolicyMaxClaim, CoveredDiagnosis[4], EdDSA signature, MerkleSiblings[8], MerkleDirections[8]. Witness JSON marshaling errors; envelope has no witness properties; Gateway path does not log witness values.

Trust and replay boundaries

Authority/root/policy are external trust anchors. Prototype key registration checks curve, non-identity and subgroup; registry operator authorization remains external. Recipient/deployment names map to configured IDs. Symbolic protocol identifier **zerotrustblock.healthclaim.v1** maps to frozen Z7 field **202610041**; arbitrary domain identifiers are rejected. Challenge freshness, durable UsedNullifiers, expiry and revocation remain external. VerifyHealthClaimProof checks request/anchor bindings but does not atomically consume fresh challenge/nullifier state for independent envelopes.

Tests and security checks

Test	Property	Result
Valid witness -> proof -> local verify -> envelope	End-to-end proof lifecycle	PASS
Mutate all 13 public fields	Statement/context/anchor binding	PASS
Missing witness / malformed scalar	Input validation fails closed	PASS
Claim 1001 > 1000 ceiling	Claim ceiling relation	PASS
Wrong Merkle path / untrusted root version	Membership and root anchor	PASS
Policy witness mismatch / untrusted version	Policy commitment/version	PASS
Unregistered authority / wrong registered key	Trust anchor and signature	PASS
Malformed signature encoding	Reject without panic	PASS
Wrong challenge/recipient/deployment	Challenge and ID mapping	PASS
Used nullifier / consumed challenge reuse	Prototype replay state	PASS
Intentional local verify rejection	Fail closed; no envelope	PASS
Envelope/witness serialization	Private fields absent; witness refused	PASS
Wrong proof hash	Identifier matches proof bytes	PASS
Existing IPFS round-trip/tamper tests	Gateway regressions	PASS

Errors and reproducibility

Errors distinguish INPUT_ERROR, CRYPTOGRAPHIC_ERROR, TRUST_REGISTRY_ERROR, REPLAY_ERROR and INTERNAL_ERROR without witness leakage. `GOCACHE=/tmp/ztb-z10-gocache ./scripts/validate-z10.sh` passed: Go 1.25.8, Gateway tests, then 10-iteration benchmark. Separate `go mod verify` exits for missing ziphash on local replace modules `zerotrust/zkp` and `zerotrust/healthclaimhardened`; distinct from passing source tests.

Performance and remaining limits

Final Gateway-local benchmark, 10 iterations. Gnark setup occurs once outside timed iterations. Fabric, network, chaincode, Docker and IPFS latency excluded.

Measure	Observed
Witness/assignment construction	1,664 ns/op (1.664 us/op)
Proof generation	70,227,173 ns/op (70.227 ms/op)
Local Groth16 verification	847,792 ns/op (0.848 ms/op)
Total local Gateway processing	70,486,692 ns/op (70.487 ms/op); benchmark 71.497 ms/op
Allocations	21,765,937 B/op; 3,638 allocs/op
Circuit	22,021 R1CS constraints; 13 public and 31 private signals (existing Z7).
Serialized proof	164 bytes (existing Z7 BN254 Groth16 measurement).

Interpretation

Proof generation contributes about 98.2% of measured local processing. This small local run is a prototype measurement, not a throughput guarantee; it does not replace Z9.

Known limitations

1. Memory registries/challenge/UsedNullifiers lose state on restart and do not provide shared atomic replay protection.
2. Independent verifiers must enforce challenge freshness and durable replay acceptance.
3. Setup/keys are in-memory; no production ceremony, persistence, custody, rotation or cross-process VK distribution.
4. Authenticated root/policy sources and registry operator authorization remain external assumptions.
5. Gateway requires Go 1.25.8 and upgrades shared gnark dependencies; tests pass, but the former Go 1.22.2 toolchain is insufficient without auto selection.
6. No HTTP API/private-record provider; caller supplies request/witness. Fabric submission is Z11.

Outcome: Z10 implemented and tested as a Fabric-independent Gateway ZKP library. PASS WITH OBSERVATIONS. STOPPED BEFORE Z11.